Customer Shopping Behaviour Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
    - Customer demographics (Age, Gender, Location, Subscription Status)
    - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
    - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

● Data Loading: Imported the dataset using pandas.

● Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

● Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

● Column Standardization: Renamed columns to snake case for better readability and documentation.

● Feature Engineering:

- ○ Created age_group column by binning customer ages.
- ○ Created purchase_frequency_days column from purchase data.

● Data Consistency Check: Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

● Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

Result Grid | Filter Rows:

gender	total_revenue_in_USD
Male	157890
Female	75191

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

Result Grid | Filter Rows:

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91

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3. Top 5 Products by Rating – Found products with the highest average review ratings.

Result Grid | Filter Rows:

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

Result Grid | Filter Rows:

shipping_type	avg_purchase
Express	60.4752
Standard	58.4602

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subscription_status	total_customer	total_revenue	average_spend
Yes	1053	62645	59.49
No	2847	170436	59.87

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

Result Grid | Filter Rows:

item_purchased	discount_percentage
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

49.66%

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

-- We can also calculate the segment directly inside the GROUP BY or by using CTEs.

Result Grid | Filter Rows:

customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

8. Top 3 Products per Category – Listed the most purchased products within each category.

Result Grid | Filter Rows: | Ex

category	item_purchased	purchase_count
Accessories	Jewelry	171
Accessories	Sunglasses	161
Accessories	Belt	161
Clothing	Blouse	171
Clothing	Pants	171
Clothing	Shirt	169
Footwear	Sandals	160
Footwear	Shoes	150
Footwear	Sneakers	145
Outerwear	Jacket	163
Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

Result Grid | Filter Rows:

subscription_status	repeat_buyer
Yes	958
No	2518

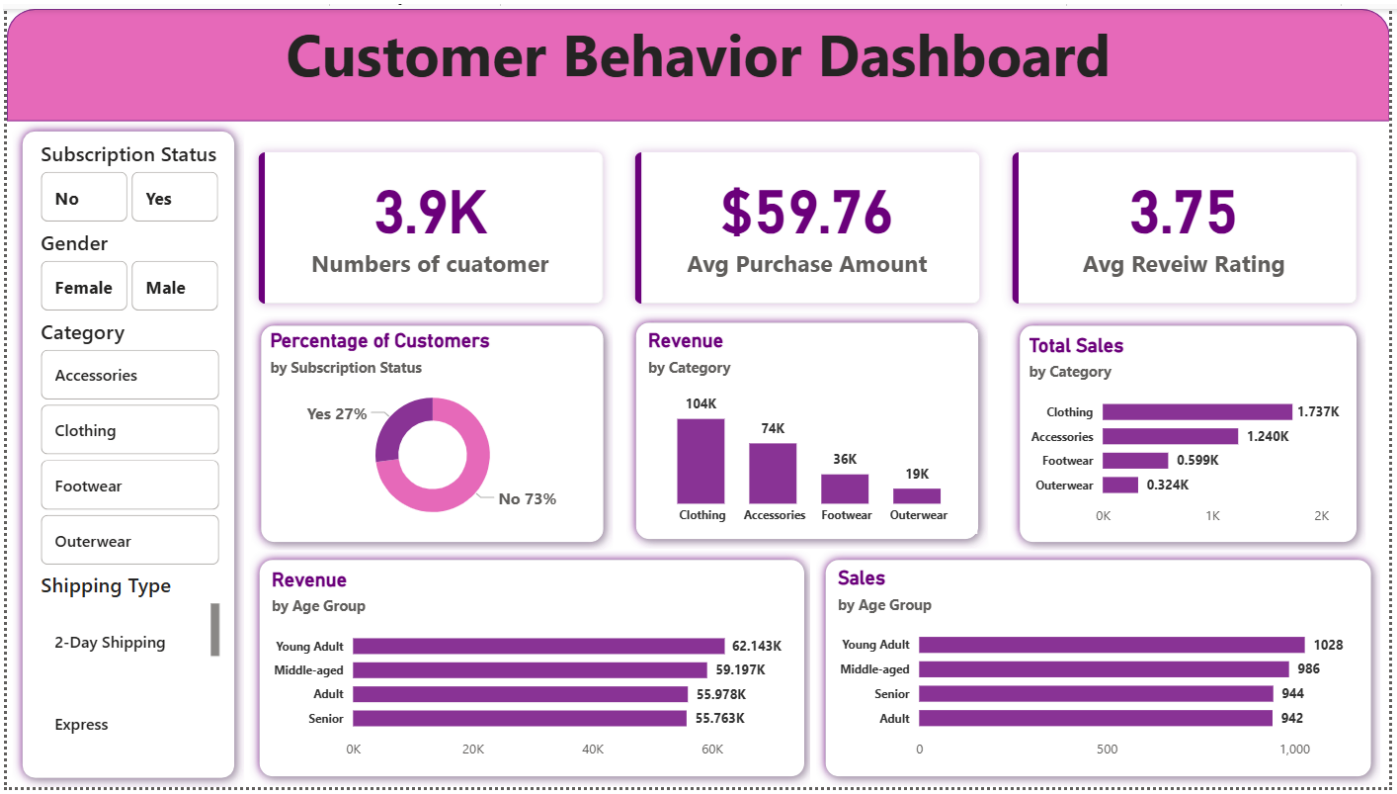
10. Revenue by Age Group – Calculated total revenue contribution of each age group.

Result Grid | Filter Rows:

age_group	revenue_contribution
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

● Boost Subscriptions – Promote exclusive benefits for subscribers.

● Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.

● Review Discount Policy – Balance sales boosts with margin control.

● Product Positioning – Highlight top-rated and best-selling products in campaigns.

● Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.